

REPORT FOR OPEN LAB PROJECT
ANALYZING THE LIFT AND DRAG FORCE RELATION FOR A CAR SPOILER WITH
VARIABLE ANGLE-OF-ATTACK

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Submitted to: Course instructor (name omitted for privacy)

1 INTRODUCTION

Rear spoilers are aerodynamic elements added to race cars that increase downforce to improve tire normal force which can aid in cornering acceleration. A spoiler works in the opposite fashion to an airplane wing that produces lift but is inverted to utilize the lift to push down on the car. It is important to optimize the rear wing such that you maximize the downforce it can generate while not undergoing boundary separation. Boundary separation occurs when the air detaches from the airfoil and turns into turbulent flow. By increasing the Angle of Attack (AoA), you can increase the lift forces, with the tradeoff of increasing the drag force. Determining the point at which the wing is operating near levels of boundary separation ensures that the user will not experience diminishing returns when increasing the angle of attack to get more downforce.

In this lab, a test model with incremented AoA points on wing endplates will be adjusted to analyze the boundary separation behavior of a wing on a race car. Lift and drag force can be estimated with the lift and drag equations with the coefficient of lift and coefficient of drag respectively as seen below.

$$F_l = \frac{1}{2} \rho V^2 * A * C_l \quad \text{Eq. 1}$$

$$F_d = \frac{1}{2} \rho V^2 * A * C_d \quad \text{Eq. 2}$$

Where the variable carries the following meaning:

F = Force, Lift or drag

ρ = Air Density

A = Surface area of wing sections

C_l/C_d = Coefficient of lift and drag

2 OBJECTIVE

The objective of this project is to determine the relationship between lift and drag of a vehicle fitted with a spoiler while varying angle of attack at a constant air velocity. The flow separation point will be determined from the lift vs. angle of attack relationship and will be used as the optimal point for maximizing downforce while minimizing drag. The lift and drag coefficients as a function of the angle of attack will also be determined.

3 METHOD

3.1 Experiment Facility

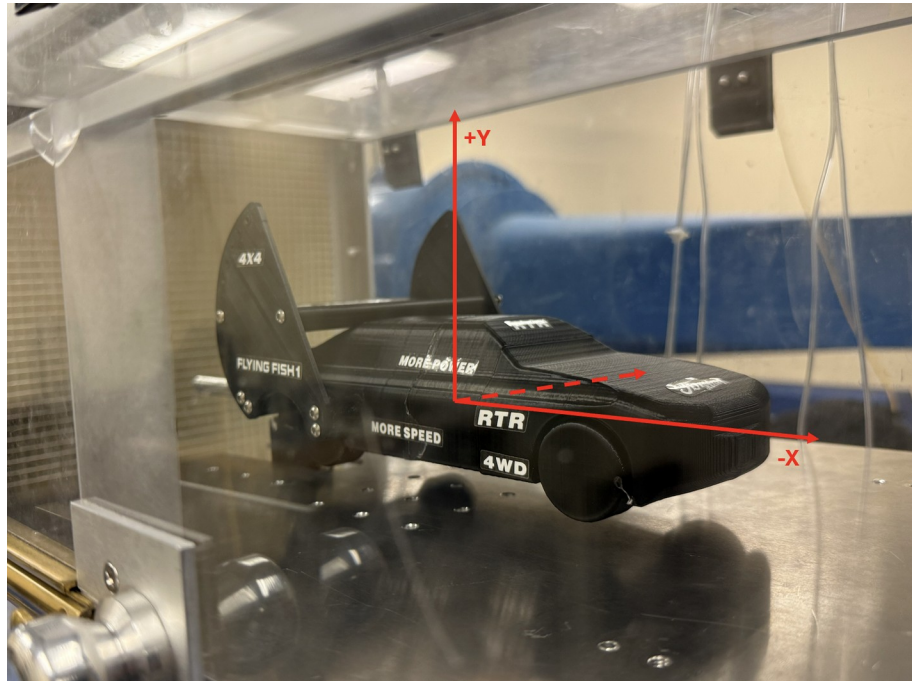


Fig. 3.1 Model Placed in Test Section

The wind tunnel used in this lab is a scaled down version of common wind tunnels used but operates on the same premise for a low-speed open-circuit wind tunnel. Air first enters the intake and honeycomb which undergoes a contraction to increase the wind velocity and ensure the laminar flow of the air entering the test section. Air leaving the test section undergoes an expansion in the diffuser to reduce the wind velocity to reduce energy consumption by the fan placed at the exit which drives the airflow all the way through the intake. Origin is placed at the center of the cylinder, and measurement occurs down the central plane only assuming no variation along the length. Positive X direction is pointing in-line with the free stream wind velocity downwind while positive Y is pointing up towards the top of the test section.

The present lab is done in a wind tunnel located at the ME308 lab, which is a low-speed open-circuit wind tunnel (ELD, Model 402B, **Fig. 3.2**). The dimensions of its test section are 30.5 cm (width) $\times$ 30.5 cm (height) $\times$ 61 cm (length). It can be operated at a range of wind speeds from 3.0 to 48.7 m/s. A centrifugal belt-driven fan with a 12.40-inch wheel diameter is driven by a 10 HP motor to move the air. A variable frequency drive (VFD) is used to adjust the rotating of the fan at different speeds and thus adjusts the wind speed in the test section.

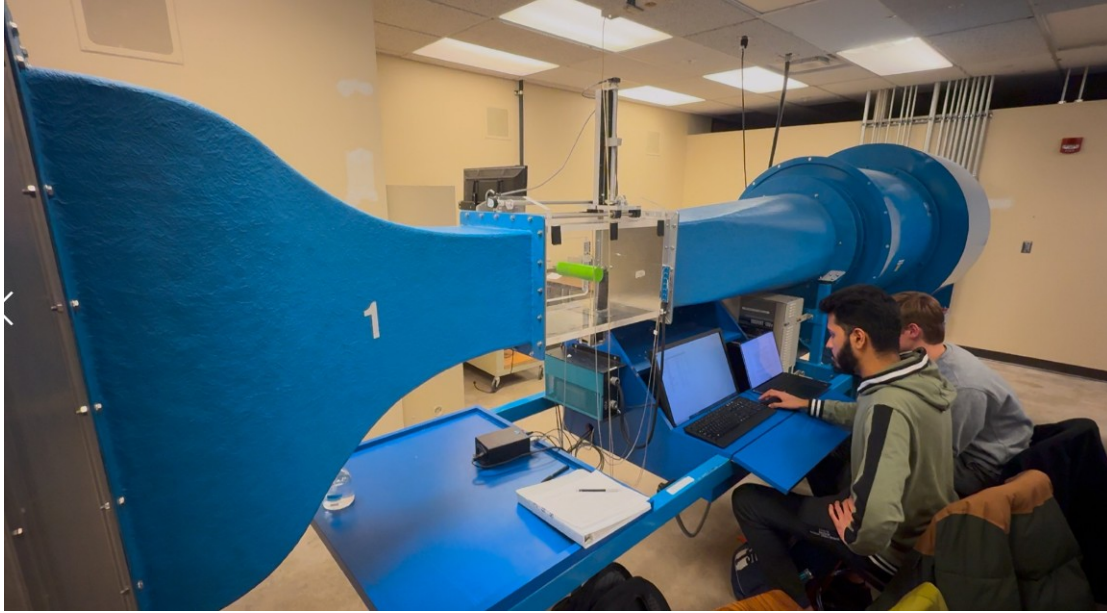


Fig. 3.2 Wind tunnel used in open lab

3.2 Test Model

The test model is based on a race car sedan which is outfitted with a rear wing including adjustment from 0 to 60 degrees, in increments of 10 degrees. The 10-degree increment was chosen due to geometric constraints with the close spacing of the M3 bolt holes. The model was 3D printed using PLA plastic, to ensure high accuracy of the test specimen. A central bolt is used as the main pivot point in the airfoil section as seen in blue. The end plates are fastened the car body with more fasteners to ensure structural integrity at high speeds.

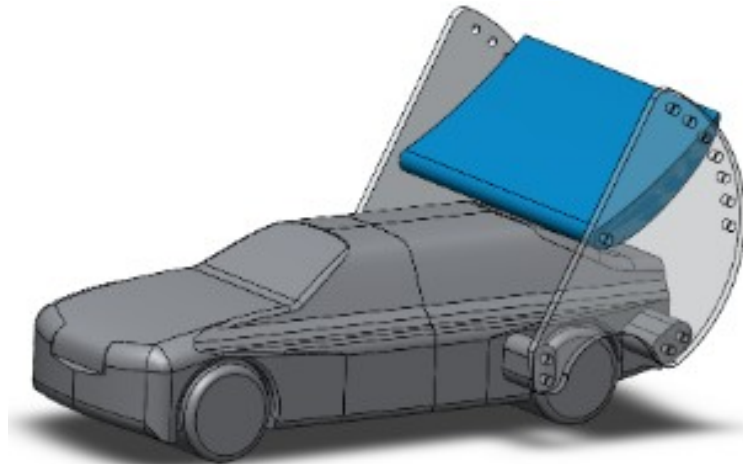


Fig. 3.3 Isometric View of Car Model

As seen in Appendix B.1, $\frac{3}{8}$ " holes were included in the rear of the car to attach to the load cell; it was placed at the very bottom of the model not to not impede the flow around the airfoil section. A large pocket is placed in the center of the car which allows for wrench clearance with the all-thread. The wing was able to be adjusted in CAD to check for clearances or interferences before

manufacturing began. The general dimension of the model is 8” length, 3” width, and 4” height, which is able to be made on a desktop 3D printer by splitting up the main body into two sections as seen by the split line in Fig. 3.3

3.3 Instrumentation

The force dynamometer measures lift and drag forces exerted onto the test model through the deflection of two pairs of restrained cantilevered beams separated by spacer blocks. Under the small deflection assumption, the deflection of these beams is proportional to the forces exerted onto the part. The deflection of the part is detected by the linear variable differential transform (LVDT), which converts the mechanical motion of the beams to electrical signals. These electrical signals are amplified with electrical circuits. The dynamometer can measure lift and drag from 0 – 100N. The relationship between Force and Voltage is described below in **Eq. 1**,

$$F = K * [(V_{total} - V_{0,offset,total}) - (V_{ts} - V_{0,offset,ts})] \quad \text{Eq. 3}$$

where the variables carry the following meaning:

F = Modified force (lift or drag force on test model alone) at test condition (N)

V_{total} = Reading of total voltage at test condition (with test model and test stand) (V)

$V_{0,offset,total}$ = Voltage reading at zero wind speed with a test model and test stand (V)

V_{ts} = Voltage reading with test stand only at test condition (V)

$V_{0,offset,ts}$ = Voltage reading at zero wind speed with stand-only (V)

K = Calibration Constant (note: different values for lift and drag)

3.4 Experimental Conditions

The wind speed is kept constant in this experiment, with the angle of attack being altered. One important aspect about the size of the model is that the wing surface shall be large enough that it will generate measurable changes in lift and drag. The car model is installed at the center of the wind tunnel with a ground plane; it is assumed to be unaffected by the walls of the tunnel as well as the threaded rod located directly behind the model. Equations for the lift and drag generated by just the wing are seen below in Eq 4 and 5 respectively

$$F_{Lift} = \frac{1}{2} \rho A V^2 * C_L \quad \text{Eq. 4}$$

$$F_{Drag} = \frac{1}{2} \rho A V^2 * C_d \quad \text{Eq. 5}$$

In this equation ρ represents the density of the oncoming air, A represents the projected frontal area of the wing, and C_d / C_L represents the drag/lift coefficient, which we assume to be 0.3 for these approximations. From the expected values below, the lift and drag force for the most extreme wing positions will generate changes that are measurable to the LVDT system.

$$R_e = \frac{\rho VL}{u} \quad \text{Eq. 6}$$

In this equation R_e represents the Reynold's Number for the system, V represents the speed of the oncoming air, L represents the chord length of the airfoil, u represents the dynamic viscosity of the fluid, and ρ represents the density of the air. The figure below provides additional information regarding wind speeds, test specimens, and Reynolds Numbers.

Table 3.1 Experimental Conditions Summary

AoA (Degrees)	Frontal Area (m²)	Flow speed (m/s)	Lift Force Expected (N)	Drag Force Expected (N)	Reynolds #
0	0.00327	30.880 m/s	-0.7	2.2	106,155
60	0.00822	30.880 m/s	-4.2	6.2	106,155

3.5 Procedures

First, the load cell was calibrated without the car model attached, next, the main test began by placing the car model in the wind tunnel section without the rear wing installed. After the fasteners were tightened the wind tunnel was set to 40Hz wind speed setting. Next the wing was installed at an angle of attack of 0 degrees with respect to the chord line, and the lift and drag was again measured for 3 trials. This was then repeated in increments of 10 degrees until reaching the max angle of attack (max angle of attack still undetermined). The wind speed will be held constantly to simplify the test and ensure it can be completed withing the allotted time range.

Test Conditions

- 40 Hz frequency setting which corresponds to 30.880 m/s wind speed.
- Angle of attack 0-60 degrees

4 RESULTS

The data acquired from LabView was voltage data for drag and lift at 40 Hz at angles of attack of 0°, 10°, 20°, 30°, 40°, 50°, 60°, and the test stand only at 40 Hz and 0 Hz. Based on the voltage readings, the following equations were used to calculate the drag force (F_{drag}), and lift force (F_{lift}) for each data set:

$$F_{drag} = K_{drag} \times [(V_{total} - V_{0,offset,total}) - (V_{ts} - V_{0,offset,ts})] \quad \text{Eq. 7}$$

$$F_{lift} = K_{lift} \times [(V_{total} - V_{0,offset,total}) - (V_{ts} - V_{0,offset,ts})] \quad \text{Eq. 8}$$

The force values were calculated for voltage data over 15 seconds for each angle of attack, and the force values were averaged, and standard deviation was also calculated for the error bars.

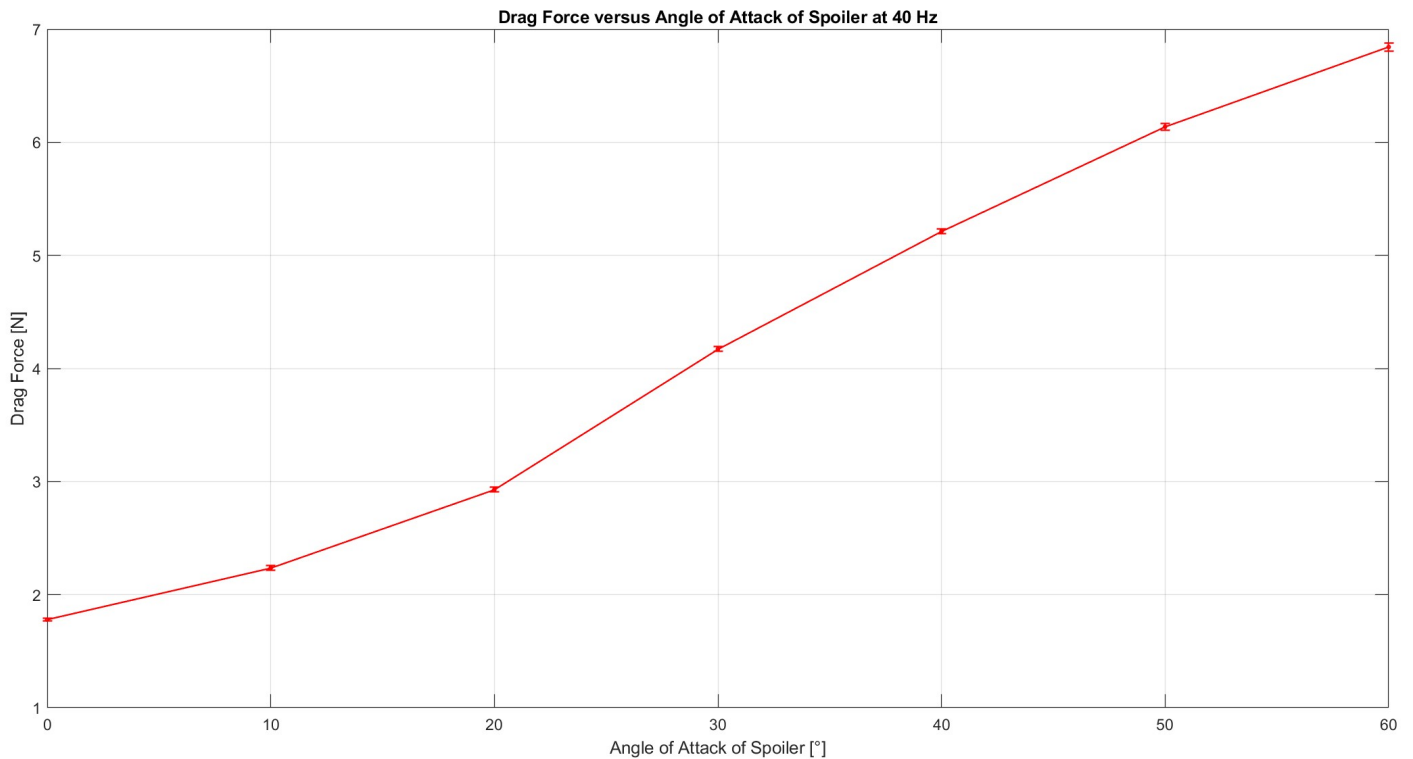


Fig. 4.1 Drag Force versus Angle of Attack of Spoiler at 40 Hz

Fig. 4.1 shows the average drag force over 15 seconds at each angle of attack. As seen, the drag force increases as the angle of attack of the spoiler is increased. The error bars are very small as visible on the plot, and the error values were calculated by taking the standard deviation of the drag force values over the 15 seconds.

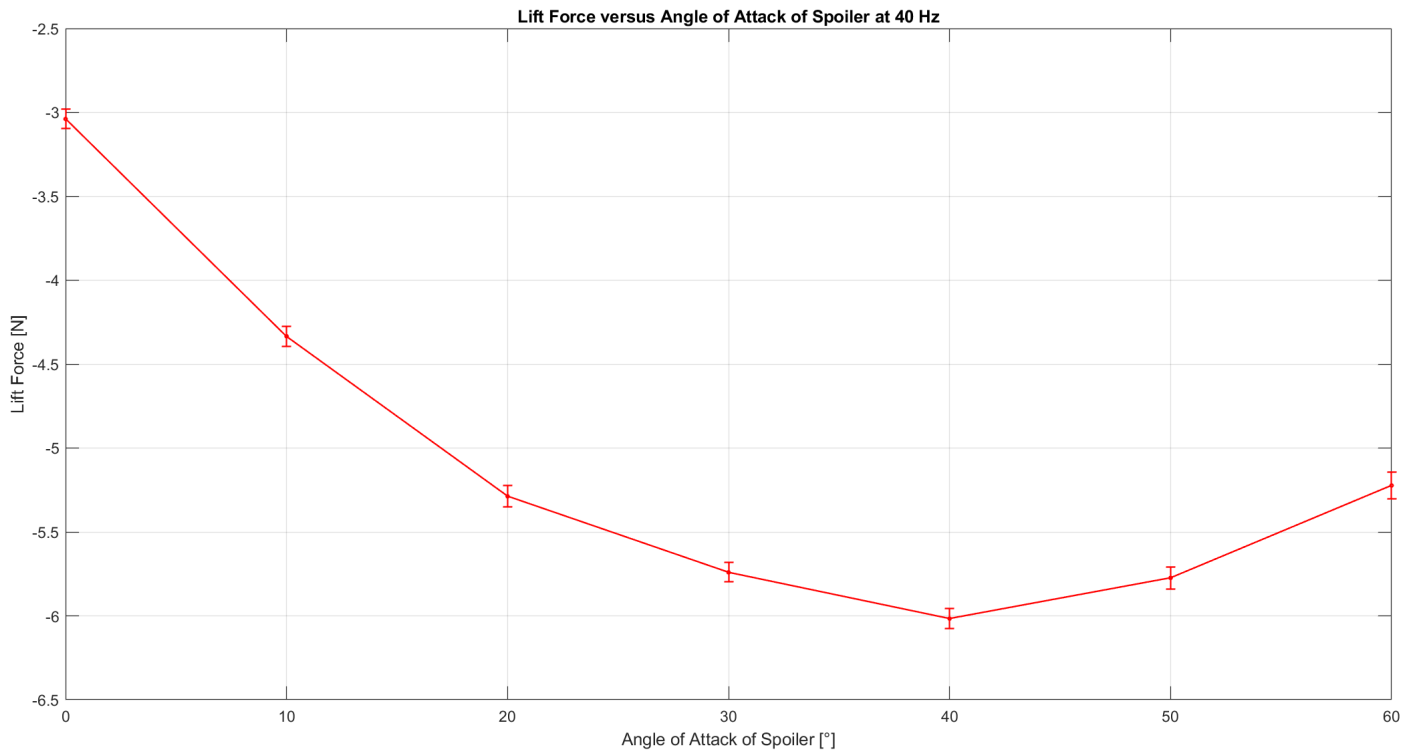


Fig. 4.2 Lift Force versus Angle of Attack of Spoiler at 40 Hz

Fig. 4.2 shows the average lift force over 15 seconds at each angle of attack. As seen, the lift force decreases, i.e., the downforce increases as the angle of attack of the spoiler is increased till 40°. As the angle of attack is increased beyond 40°, the downforce decreases. The error bars are small as visible on the plot, and the error values were calculated by taking the standard deviation of the lift force values over the 15 seconds.

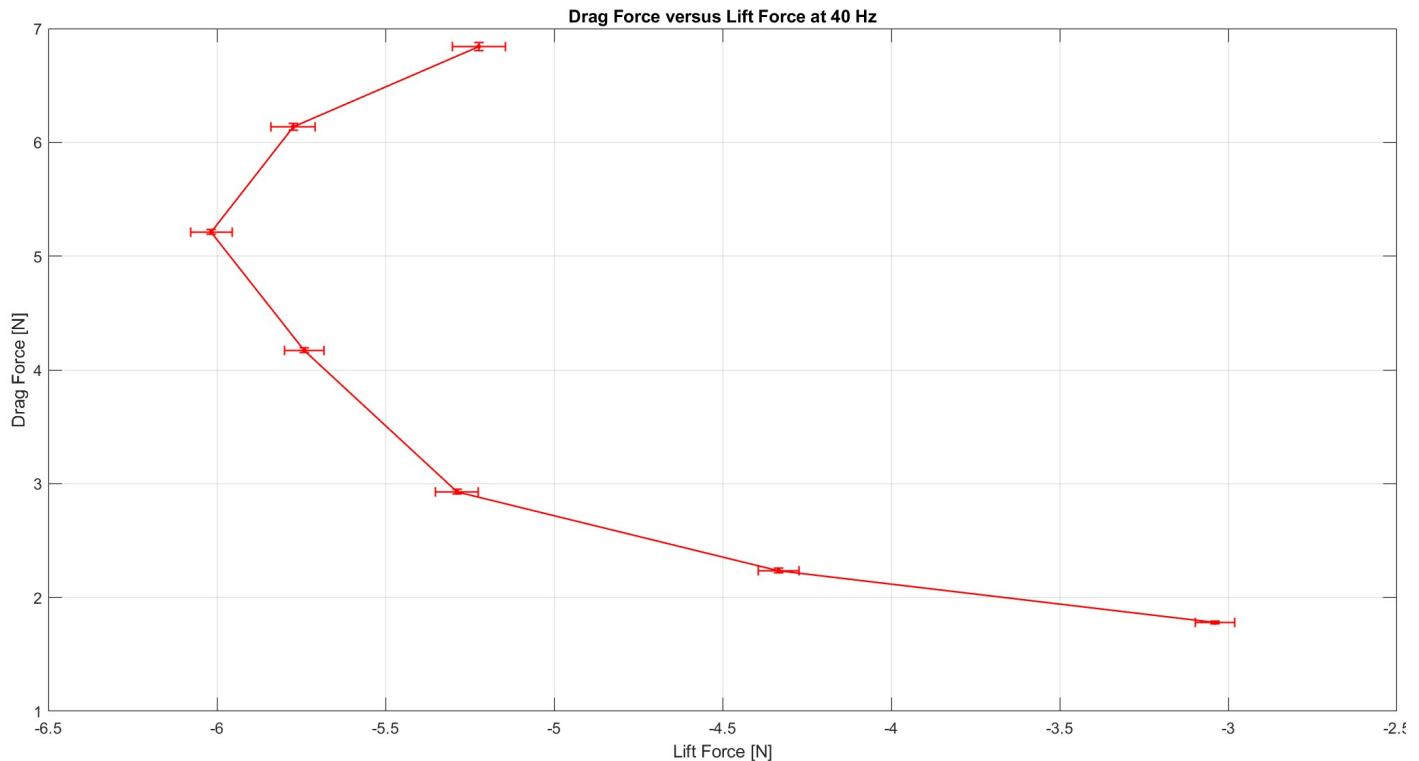


Fig. 4.3 Drag Force versus Lift Force at 40 Hz

Fig. 4.3 shows the average drag force versus the average lift force over 15 seconds at each angle of attack. As seen, the lift force decreases (downforce increases) as the drag force decreases. Beyond this point, the lift force increases (downforce decreases) as the drag force decreases further. This plot goes on to show that maximum downforce can be achieved without experiencing maximum drag force. Hence, they do not have a proportional relationship throughout. The error bars for both the drag force and lift force are small as visible on the plot, and the error values were calculated by taking the standard deviation of the force values over the 15 seconds.

5 DISCUSSION

As the Angle of Attack increases, a steady, positive trend is observed in the Drag Force. This is expected, as the higher AoA increases the frontal area of the vehicle, which has a positive, linear relationship with Drag Force. This trend remains relatively constant throughout the range of angle of attack. As the Angle of Attack increases, an initial positive trend in downforce is observed from 0 degrees to 40 degrees. Between 40-60 degrees, the downforce begins to decrease. This point is interpreted to be the flow separation point, where the airfoil stalls and ceases to produce more downforce. These results indicate that the highest downforce is possible without the highest drag. This provides an optimal angle of attack (40 degrees) for vehicle

performance, where downforce is maximized while producing maximum drag. Potential sources of error include the estimation of the frontal area and the ground plane placement. In addition, the car model was printed with a slight 2mm shift. This manufacturing defect may have influenced the flow around the body of the test vehicle.

Table 5.1 Coefficient of Drag and Lift at each Angle of Attack

Angle of Attack (Degrees)	Coefficient of Drag	Coefficient of Lift
0	0.89	0.33
10	0.89	0.48
20	0.95	0.57
30	1.18	0.62
40	1.31	0.65
50	1.39	0.63
60	1.42	0.57

As shown above in Table 5.1, the lift coefficient increases as AoA increases up to 40 degrees Angle of Attack. However, the lift coefficient decreases with subsequent increases in Angle of Attack. This is due to flow separation. This occurs due to the boundary layer conditions, where the no-slip boundary condition constrains the flow to have a zero relative velocity to the surface of the model, and the edge of the boundary layer velocity must equal the free stream velocity. In order to satisfy these boundary conditions, the flow reverses direction and causes a turbulent wake, reducing the lift coefficient.

The major limitation of this experiment is conducting the analysis of lift and drag at only one air speed point, 40Hz. The flow separation point may occur at a different Angle of Attack for different wind tunnel speeds. In addition, a simplified vehicle model was used, which may not capture the entire flow behavior around complex surfaces like a wing, intake, and underbody. Other real-world considerations include cross winds, different yaw angles, and suspension movement, which all contribute to the aerodynamic performance of a vehicle.

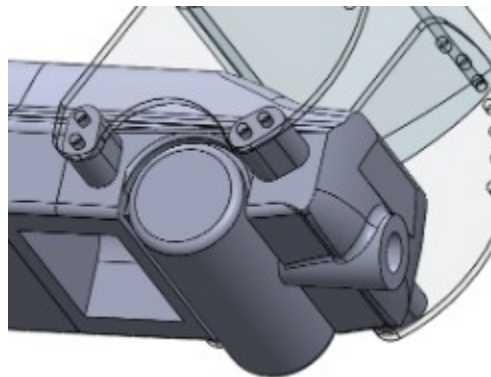
The objective was achieved – the relationship between both lift and drag as a function of airfoil AoA are displayed in Figures 4.1-4.3. A maximum downforce was observed at an AoA of 40 degrees, displaying a clear flow separation point, which serves to maximize downforce while minimizing drag. The drag and lift coefficient relationships are shown in Table 5.1, which follow similar trends to the force relationships.

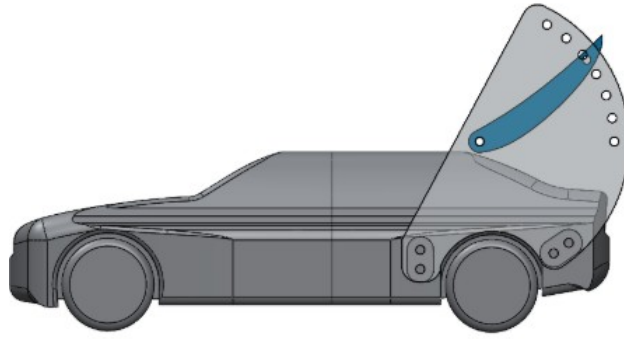
REFERENCES

1. Fox, R. W., McDonald, A. T., & Mitchell, J. W. (2020). Fox and McDonald's introduction to fluid mechanics. John Wiley & Sons.
2. Anderson, J. D. (2005). Ludwig Prandtl's boundary layer. Physics Today, 58(12), 42-48.

APPENDIX A: CONTRIBUTION OF TEAM MEMBERS

Item\Name	Binstock, Aaron	Iyer, Soumil	Johansson, Victor	Subtotal
Project Idea	33%	33%	34%	100%
Proposal Preparation	33%	33%	34%	100%
Lab Test	33%	33%	34%	100%
Data Processing		100%		100%
Report Writing				
Introduction	100%			100%
Objective			100%	100%
Method	100%			100%
Results		100%		100%
Discussion	15%		85%	100%
Edit & proofread	33%	33%	34%	100%
PPT preparation	33%	33%	34%	100%

APPENDIX B: MODEL VIEWS*B.1: Bottom Rear View*

*B.2: Model Side View***APPENDIX C: DATA PROCESSING CODE**

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clc;
clear;

data_40Hz_angle1 = [mean(readmatrix('40Hz_angle1_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle1_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_angle2 = [mean(readmatrix('40Hz_angle2_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle2_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_angle3 = [mean(readmatrix('40Hz_angle3_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle3_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_angle4 = [mean(readmatrix('40Hz_angle4_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle4_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_angle5 = [mean(readmatrix('40Hz_angle5_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle5_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_angle6 = [mean(readmatrix('40Hz_angle6_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle6_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_angle7 = [mean(readmatrix('40Hz_angle7_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_angle7_1.xlsx', 'Range', 'C:C'), 'omitnan')];

data_40Hz_no_wing = [mean(readmatrix('40Hz_model_no_wing_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_model_no_wing_1.xlsx', 'Range', 'C:C'), 'omitnan')];
data_40Hz_ts_only = [mean(readmatrix('40Hz_test_stand_only_1.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('40Hz_test_stand_only_1.xlsx', 'Range', 'C:C'), 'omitnan')];

data_0Hz_no_wing = [mean(readmatrix('0Hz_model_no_wing.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('0Hz_model_no_wing.xlsx', 'Range', 'C:C'), 'omitnan')];
data_0Hz_ts_only = [mean(readmatrix('0Hz_test_stand_only.xlsx', 'Range', 'B:B'), 'omitnan'), mean(readmatrix('0Hz_test_stand_only.xlsx', 'Range', 'C:C'), 'omitnan')];

stdev_40Hz_angle1 = [std(readmatrix('40Hz_angle1_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle1_1.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_40Hz_angle2 = [std(readmatrix('40Hz_angle2_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle2_1.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_40Hz_angle3 = [std(readmatrix('40Hz_angle3_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle3_1.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_40Hz_angle4 = [std(readmatrix('40Hz_angle4_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle4_1.xlsx', 'Range', 'C:C'), 'omitnan')];

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stdev_40Hz_angle5 = [std(readmatrix('40Hz_angle5_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle5_1.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_40Hz_angle6 = [std(readmatrix('40Hz_angle6_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle6_1.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_40Hz_angle7 = [std(readmatrix('40Hz_angle7_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_angle7_1.xlsx', 'Range', 'C:C'), 'omitnan')];

stdev_40Hz_no_wing = [std(readmatrix('40Hz_model_no_wing_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_model_no_wing_1.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_40Hz_ts_only = [std(readmatrix('40Hz_test_stand_only_1.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('40Hz_test_stand_only_1.xlsx', 'Range', 'C:C'), 'omitnan')];

stdev_0Hz_no_wing = [std(readmatrix('0Hz_model_no_wing.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('0Hz_model_no_wing.xlsx', 'Range', 'C:C'), 'omitnan')];
stdev_0Hz_ts_only = [std(readmatrix('0Hz_test_stand_only.xlsx', 'Range', 'B:B'), 'omitnan'), std(readmatrix('0Hz_test_stand_only.xlsx', 'Range', 'C:C'), 'omitnan')];

K_lift = 80.856;
K_drag = 11.196;

P_air = 766.8 * 133.322; % Pressure of air in Pa calculated from observed mmHg in the room
R_air = 287; % Unit: J/kg.K
T_air = 20.3 + 273.15; % Temperature of air in K observed in the room
rho = P_air / (R_air * T_air); % Density of air in kg/m^3 in the room

V_40Hz = 30.880;

F_40Hz_angle1_lift = K_lift * ((data_40Hz_angle1(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));
F_40Hz_angle2_lift = K_lift * ((data_40Hz_angle2(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));
F_40Hz_angle3_lift = K_lift * ((data_40Hz_angle3(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));
F_40Hz_angle4_lift = K_lift * ((data_40Hz_angle4(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));
F_40Hz_angle5_lift = K_lift * ((data_40Hz_angle5(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));
F_40Hz_angle6_lift = K_lift * ((data_40Hz_angle6(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));
F_40Hz_angle7_lift = K_lift * ((data_40Hz_angle7(1) - data_0Hz_no_wing(1)) - (data_40Hz_ts_only(1) - data_0Hz_ts_only(1)));

F_40Hz_angle1_drag = K_drag * ((data_40Hz_angle1(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));
F_40Hz_angle2_drag = K_drag * ((data_40Hz_angle2(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));
F_40Hz_angle3_drag = K_drag * ((data_40Hz_angle3(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));
F_40Hz_angle4_drag = K_drag * ((data_40Hz_angle4(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));
F_40Hz_angle5_drag = K_drag * ((data_40Hz_angle5(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));
F_40Hz_angle6_drag = K_drag * ((data_40Hz_angle6(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));
F_40Hz_angle7_drag = K_drag * ((data_40Hz_angle7(2) - data_0Hz_no_wing(2)) - (data_40Hz_ts_only(2) - data_0Hz_ts_only(2)));

error_40Hz_angle1_lift = K_lift * ((stdev_40Hz_angle1(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));
error_40Hz_angle2_lift = K_lift * ((stdev_40Hz_angle2(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));
error_40Hz_angle3_lift = K_lift * ((stdev_40Hz_angle3(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));
error_40Hz_angle4_lift = K_lift * ((stdev_40Hz_angle4(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));
error_40Hz_angle5_lift = K_lift * ((stdev_40Hz_angle5(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));
error_40Hz_angle6_lift = K_lift * ((stdev_40Hz_angle6(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));
error_40Hz_angle7_lift = K_lift * ((stdev_40Hz_angle7(1) - stdev_0Hz_no_wing(1)) - (stdev_40Hz_ts_only(1) - stdev_0Hz_ts_only(1)));

error_40Hz_angle1_drag = K_drag * ((stdev_40Hz_angle1(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));
error_40Hz_angle2_drag = K_drag * ((stdev_40Hz_angle2(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));
error_40Hz_angle3_drag = K_drag * ((stdev_40Hz_angle3(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));
error_40Hz_angle4_drag = K_drag * ((stdev_40Hz_angle4(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));
error_40Hz_angle5_drag = K_drag * ((stdev_40Hz_angle5(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));
error_40Hz_angle6_drag = K_drag * ((stdev_40Hz_angle6(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));
error_40Hz_angle7_drag = K_drag * ((stdev_40Hz_angle7(2) - stdev_0Hz_no_wing(2)) - (stdev_40Hz_ts_only(2) - stdev_0Hz_ts_only(2)));

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angle = [0, 10, 20, 30, 40, 50, 60];
F_lift = [F_40Hz_angle1_lift, F_40Hz_angle2_lift, F_40Hz_angle3_lift, F_40Hz_angle4_lift, F_40Hz_angle5_lift, ✓
F_40Hz_angle6_lift, F_40Hz_angle7_lift];
F_drag = [F_40Hz_angle1_drag, F_40Hz_angle2_drag, F_40Hz_angle3_drag, F_40Hz_angle4_drag, F_40Hz_angle5_drag, ✓
F_40Hz_angle6_drag, F_40Hz_angle7_drag];

error_lift = [error_40Hz_angle1_lift, error_40Hz_angle2_lift, error_40Hz_angle3_lift, error_40Hz_angle4_lift, ✓
error_40Hz_angle5_lift, error_40Hz_angle6_lift, error_40Hz_angle7_lift];
error_drag = [error_40Hz_angle1_drag, error_40Hz_angle2_drag, error_40Hz_angle3_drag, error_40Hz_angle4_drag, ✓
error_40Hz_angle5_drag, error_40Hz_angle6_drag, error_40Hz_angle7_drag];

figure (1);
errorbar(F_lift, F_drag, error_drag, error_drag, error_lift, error_lift, '-ro', 'LineWidth', 1, 'MarkerSize', 2);
xlabel('Lift Force [N]');
ylabel('Drag Force [N]');
title('Drag Force versus Lift Force at 40 Hz');
grid on;

figure (2);
errorbar(angle, F_lift, error_lift, '-ro', 'LineWidth', 1, 'MarkerSize', 2);
xlabel('Angle of Attack of Spoiler [°]');
ylabel('Lift Force [N]');
title('Lift Force versus Angle of Attack of Spoiler at 40 Hz');
grid on;

figure (3);
errorbar(angle, F_drag, error_drag, '-ro', 'LineWidth', 1, 'MarkerSize', 2);
xlabel('Angle of Attack of Spoiler [°]');
ylabel('Drag Force [N]');
title('Drag Force versus Angle of Attack of Spoiler at 40 Hz');
grid on;

```